

## Inverse of a Matrix

### Definition

Let  $A$  be an  $n \times n$  matrix. Then  $B$  is an **inverse** of  $A$  if and only if  $AB = I_n$  and  $BA = I_n$ . What does this imply about the size of  $B$ ?

Let us assume  $B$  is an  $p \times k$  matrix, then for  $AB$  to be defined, we have  $p = n$ .  
Also for  $BA$  to be defined, we have  $k = n$ , which means  $B$  is an  $n \times n$  matrix.

### Theorem (Uniqueness of an Inverse)

If  $A$  is a square matrix and  $B$  and  $C$  are inverses of  $A$ , then  $B = C$ .

$$B \text{ and } C \text{ are inverses of } A \left\{ \begin{array}{l} BA = AB = I \\ CA = AC = I \end{array} \right\} \Rightarrow B = BI = B(AC) = (BA)C = IC = C$$

## Definitions

Let  $A$  be a square matrix.

- The inverse of  $A$ , if it exists, is denoted by  $A^{-1}$ , and

$$AA^{-1} = I = A^{-1}A$$

- If  $A$  has an inverse, then we say that  $A$  is **invertible**.

## The inverse of a $2 \times 2$ matrix

Suppose  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and  $ad - bc \neq 0$ , then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

**In HW5, students will prove that the inverse of a 2x2 matrix is in the form above.**

Example: Show that  $A = \begin{bmatrix} 0 & k \\ 0 & k \end{bmatrix}$ , where  $k \in \mathbb{R}$ , does not have an inverse.

If  $A^{-1}$  exists, then

$$A A^{-1} = \begin{bmatrix} 0 & k \\ 0 & k \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} kc & kd \\ kc & kd \end{bmatrix} = k \begin{bmatrix} c & d \\ c & d \end{bmatrix} \quad \text{which cannot be } I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

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## Questions:

Suppose that  $A$  is any  $n \times n$  matrix.

- How do we know whether or not  $A^{-1}$  exists?
- If  $A^{-1}$  exists, how do we find it? *We use the matrix inversion algorithm.*

## The Matrix Inversion Algorithm

Let  $A$  be an  $n \times n$  matrix. To find  $A^{-1}$ , if it exists,

- take the  $n \times 2n$  matrix

$$[A \mid I_n]$$

obtained by augmenting  $A$  with the  $n \times n$  identity matrix,  $I_n$ .

- Perform elementary row operations to transform  $[A \mid I_n]$  into a reduced row-echelon matrix.

*Example:  $A = \begin{bmatrix} 2 & -7 \\ 5 & -18 \end{bmatrix}$ , find  $A^{-1}$ .*

$$\begin{aligned} [A \mid I_n] &= \left[ \begin{array}{cc|cc} 2 & -7 & 1 & 0 \\ 5 & -18 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{cc|cc} 2 & -7 & 1 & 0 \\ 1 & -4 & -2 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{cc|cc} 1 & -4 & -2 & 1 \\ 2 & -7 & 1 & 0 \end{array} \right] \\ &\rightarrow \left[ \begin{array}{cc|cc} 1 & -4 & -2 & 1 \\ 0 & 1 & 5 & -2 \end{array} \right] \rightarrow \left[ \begin{array}{cc|cc} 1 & 0 & 18 & -7 \\ 0 & 1 & 5 & -2 \end{array} \right] = [I_2 \mid A^{-1}] \end{aligned}$$

Note that for a  $2 \times 2$  matrix, we can:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{-1} \begin{bmatrix} -18 & 7 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 18 & -7 \\ 5 & -2 \end{bmatrix}$$

Although the above formula for  $\bar{A}^{-1}$  is much quicker for  $2 \times 2$  matrices, as we will see later, the general formula for  $n \times n$  matrices ( $n \geq 3$ ) is less efficient and more complicated to use when compared to the matrix inversion algorithm.

## Theorem (Matrix Inverses)

Let  $A$  be an  $n \times n$  matrix. Then the following conditions are equivalent.

- 1  $A$  is invertible.
- 2 the reduced row-echelon form on  $A$  is  $I$ .
- 3  $[A \mid I_n]$  can be transformed into  $[I_n \mid A^{-1}]$  using the Matrix Inversion Algorithm.

Assume that  $[A \mid I_n]$  can be transformed into  $[I_n \mid B]$  using EROs. This means the RREF of  $A$  is  $I$ . Also, if the following is possible using EROs:

$$[A \mid \vec{e}_1 \ \vec{e}_2 \ \dots \ \vec{e}_n] \longrightarrow [I_n \mid \vec{B}_1 \ \vec{B}_2 \ \dots \ \vec{B}_n]$$

Then  $[A \mid \vec{e}_i] \longrightarrow [I_n \mid \vec{B}_i]$  is possible, which means

$A\vec{x}_i = \vec{e}_i$  has the solution  $\vec{x}_i = \vec{B}_i$  or  $A\vec{B}_i = \vec{e}_i$ , hence

$$AB = [A\vec{B}_1 \ A\vec{B}_2 \ \dots \ A\vec{B}_n] = [\vec{e}_1 \ \vec{e}_2 \ \dots \ \vec{e}_n] = I.$$

Later, we will show that if  $AB = I$ , then  $BA = I$  and hence

$B = A^{-1}$ . (This will be shown when we study the invertible matrix theorem.)

Also, see an example of this on next page.

## Theorem (Matrix Inverses)

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- 1  $A$  is invertible.
- 2 the reduced row-echelon form on  $A$  is  $I$ .
- 3  $[A \mid I_n]$  can be transformed into  $[I_n \mid A^{-1}]$  using the Matrix Inversion Algorithm.

$$\text{Ex: } A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

$$[A \mid I_2] = \left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{array} \right] \xrightarrow{R_1 + 2R_2} \left[ \begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 0 & -1 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2} \left[ \begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & -1 \end{array} \right]$$

$$\begin{array}{l} A \vec{b}_1 = \vec{e}_1 \\ A \vec{b}_2 = \vec{e}_2 \end{array} \rightarrow AB = \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 \end{bmatrix} \\ = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 \end{bmatrix} \\ = I$$



# Problem

Let  $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$ . Find the inverse of  $A$ , if it exists.

$$\begin{aligned}
 &\left[ \begin{array}{ccc|ccc} 3 & 1 & 2 & 1 & 0 & 0 \\ 1 & -1 & 3 & 0 & 1 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 3 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 0 & 4 & -7 & 1 & -3 & 0 \\ 0 & 3 & 1 & 0 & -1 & 1 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 0 & 1 & -8 & 1 & -2 & -1 \\ 0 & 3 & 1 & 0 & -1 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 0 & 1 & -8 & 1 & -2 & -1 \\ 0 & 0 & 25 & -3 & 5 & 4 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -1 & -1 \\ 0 & 1 & -8 & 1 & -2 & -1 \\ 0 & 0 & 1 & -\frac{3}{25} & \frac{5}{25} & \frac{4}{25} \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -1 & -1 \\ 0 & 1 & 0 & \frac{1}{25} & -\frac{16}{25} & \frac{7}{25} \\ 0 & 0 & 1 & -\frac{3}{25} & \frac{5}{25} & \frac{4}{25} \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{10}{25} & 0 & -\frac{15}{25} \\ 0 & 1 & 0 & \frac{1}{25} & -\frac{16}{25} & \frac{7}{25} \\ 0 & 0 & 1 & -\frac{3}{25} & \frac{5}{25} & \frac{4}{25} \end{array} \right] = [I_3 | A^{-1}] \rightarrow A^{-1} = \begin{bmatrix} \frac{10}{25} & 0 & -\frac{15}{25} \\ \frac{1}{25} & -\frac{16}{25} & \frac{7}{25} \\ -\frac{3}{25} & \frac{5}{25} & \frac{4}{25} \end{bmatrix}
 \end{aligned}$$

## Inverse of a Matrix and the Matrix Equation $A\vec{x} = B$

### Example

Consider the system of linear equations

$$\begin{aligned} 2x - 7y &= 3 \\ 5x - 18y &= 8 \end{aligned}$$

and use the concept of **the inverse of a matrix** to solve the system.

We can write the matrix eq.  $A\vec{x} = \vec{b}$  as follows:  $A\vec{x} = \begin{bmatrix} 2 & -7 \\ 5 & -18 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 8 \end{bmatrix}}_{\vec{b}}$

If  $A$  is invertible then:  $\vec{A}^{-1}(A\vec{x}) = \vec{A}^{-1}(\vec{b})$

$$(\vec{A}^{-1}A)\vec{x} = \vec{A}^{-1}\vec{b} \rightarrow I\vec{x} = \vec{A}^{-1}\vec{b} \rightarrow \boxed{\vec{x} = \vec{A}^{-1}\vec{b}}$$

$$\vec{A}^{-1} = \begin{bmatrix} 18 & -7 \\ 5 & -2 \end{bmatrix} \rightarrow \vec{x} = \begin{bmatrix} 18 & -7 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 8 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$x = -2$  and  $y = -1$  is a unique solution for this system.

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How do we know  $\vec{x}$  is a unique solution? Let's assume that there exists more than one solution. For example, if  $\vec{w}$  is another solution then we can repeat the above process to show that  $\vec{w} = \vec{A}^{-1}\vec{b}$ . Also, since  $\vec{x} = \vec{A}^{-1}\vec{b}$ , then  $\vec{x} = \vec{w}$ , which means  $\vec{x}$  and  $\vec{w}$  are the same.

## Theorem (Inverses of Transposes and Products)

- 1 If  $A$  is an invertible matrix, then  $(A^T)^{-1} = (A^{-1})^T$ .
- 2 If  $A$  and  $B$  are invertible matrices, then  $AB$  is invertible and  $(AB)^{-1} = B^{-1}A^{-1}$ .
- 3 If  $A_1, A_2, \dots, A_k$  are invertible, then  $A_1A_2 \cdots A_k$  is invertible and  $(A_1A_2 \cdots A_k)^{-1} = A_k^{-1}A_{k-1}^{-1} \cdots A_2^{-1}A_1^{-1}$ .

- 1 If  $A$  is an invertible matrix, then  $AA^{-1} = I$ , which means

$$\begin{aligned} & (AA^{-1})^T = I^T \longrightarrow (A^{-1})^T A^T = I \\ \text{Also, since } A^{-1}A = I & \longrightarrow (A^{-1}A)^T = A^T (A^{-1})^T = I \end{aligned} \quad \left. \vphantom{\begin{aligned} & (AA^{-1})^T = I^T \longrightarrow (A^{-1})^T A^T = I \\ & (A^{-1}A)^T = A^T (A^{-1})^T = I \end{aligned}} \right\} \Rightarrow (A^{-1})^T \text{ is inverse of } A^T \text{ or} \\ & (A^{-1})^T = (A^T)^{-1}. \end{aligned}$$

Exercise: Prove ② and ③.